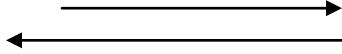


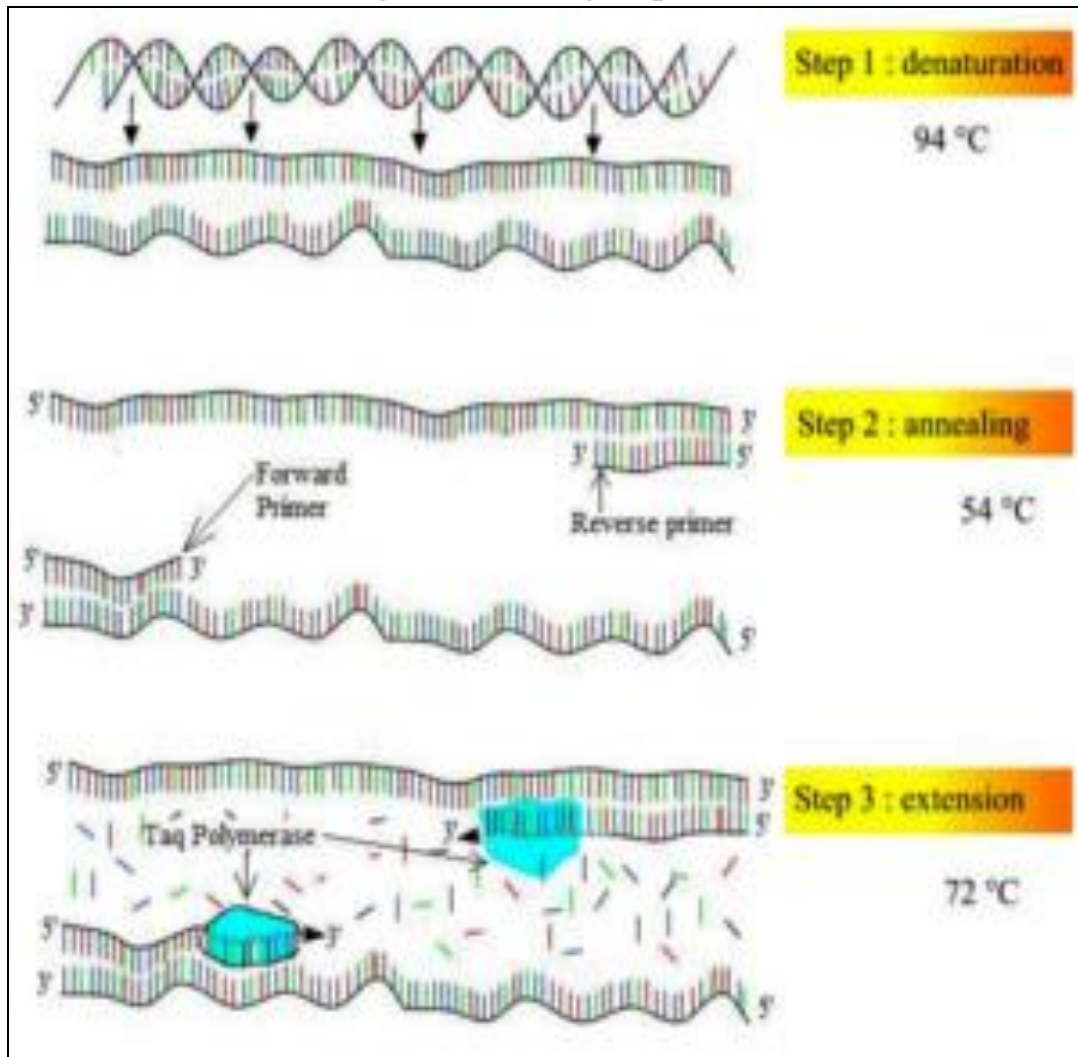
1. What is palindrome in DNA?

Ans: Palindrome in DNA is sequence of base pairs that read the same on the 2 strands when orientation of reading is kept the same.

E.g.: MALAYALAM



2. Draw a neat well labeled diagram showing steps of PCR.

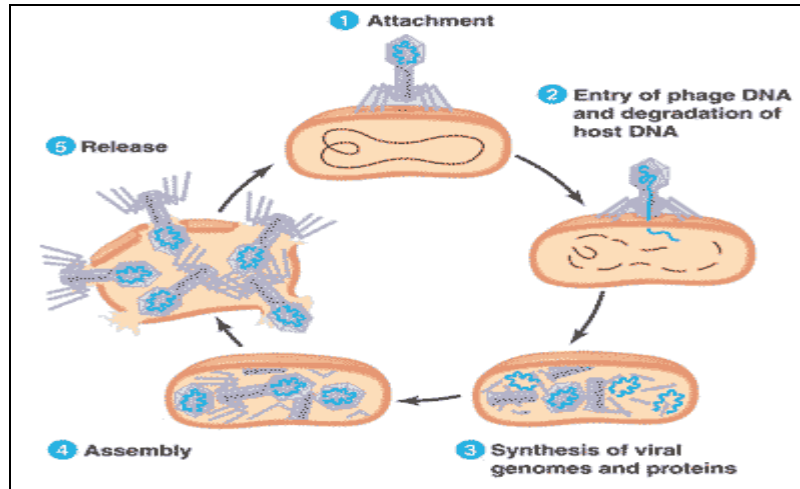


Ans:

3. What is bacteriophage? Explain lytic cycle of bacteriophage with suitable diagram?

Ans: Bacteriophage: It is a virus that infects bacteria. They replicate within bacteria & consist of outer protein capsid enclosing genetic material.

Lytic Cycle



1. Replication of bacteriophage takes place inside the specific host or bacterial cell through lytic cycle.
2. Lytic cycle is the life cycle of a phage in which progeny phages are produced & the host bacterial cell is lysed.
3. Replication takes place in following steps:-

A| Attachment

1. On the surface of bacteria, some receptors are present for attachment of bacteriophage.
2. With the right receptor, bacteriophage attaches itself to the bacteria.

B| Penetration

1. After attachment with the help of tail fibres of bacteriophage, the basal plate comes close to the surface of the bacterial cell.
2. The tail contracts & the genetic material are injected in bacterial cell through bacterial membrane.

C| Degradation of Host DNA:

1. After injecting the genetic material, the protein coat which remains outside the cell is now called as ghost.
2. Inside the bacterial cell the phage linear DNA cyclizes & gets ligated through cos ends to form a circular DNA.

D| Synthesis of viral DNA & protein coat

1. Instead of synthesis of host protein & nucleic acid, viral DNA & proteins are synthesized.
2. Many viral proteins necessary for formation of head, tail etc are synthesized.

E| Virion Assembly:

1. After synthesis of all proteins, virion assembly takes place; basal plate is assembled with tail.
2. The head capsid are constructed which further join with tail
3. The circular DNA is cleaved & packed into head of phage.

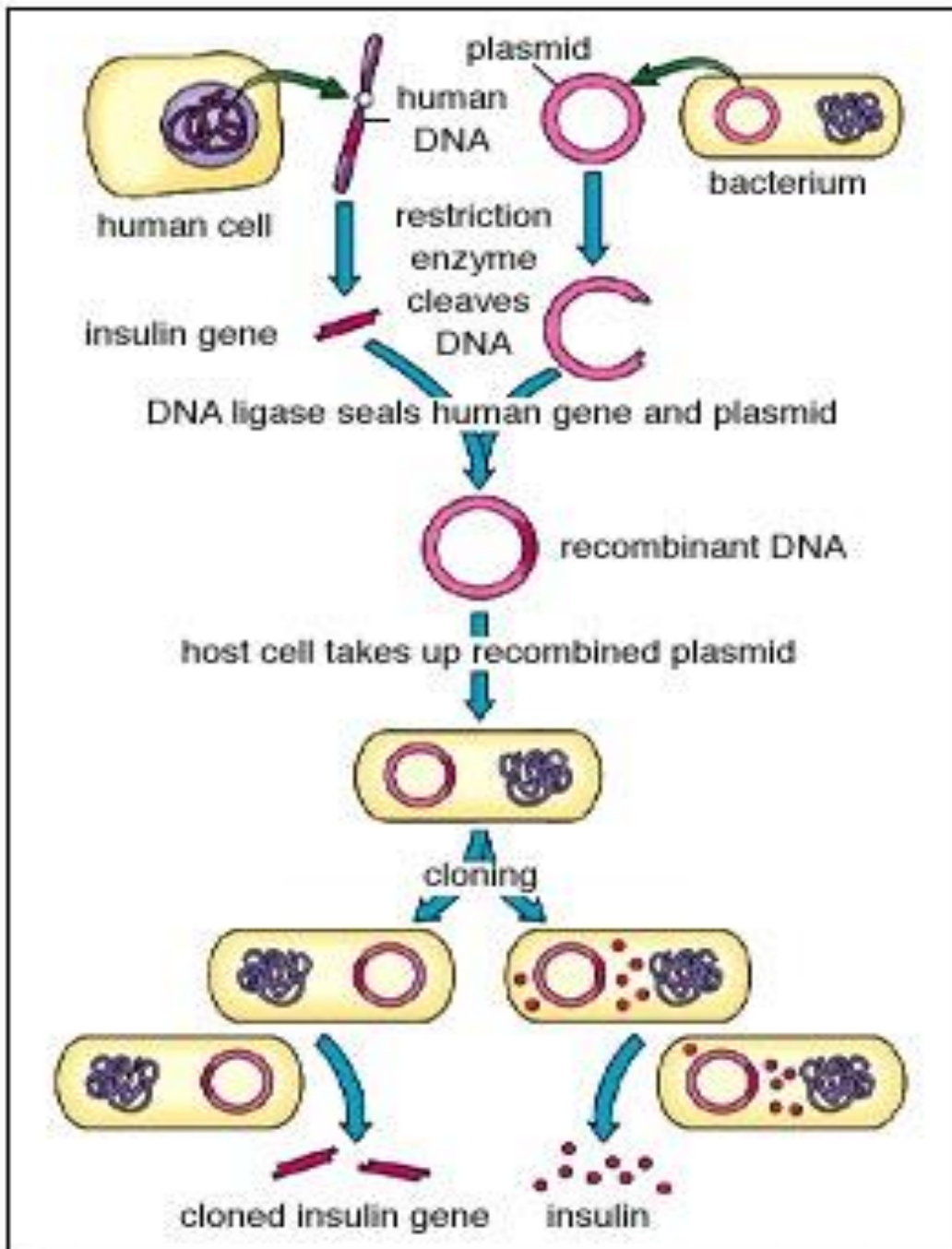
F| Release of virions

1. Within 15 minutes large numbers of phage particles are produced after the entry of phage into bacterial cell.
2. The host cell is then subjected to lysis & the phages are released.
3. It is done by an enzyme called endolysin, which breaks the cell wall of the host cell.
4. Each progeny phage particle can infect bacterial cell & produce several hundreds of phages.
5. It is estimated that by repeating the lytic cycles 4 times, a single phage can cause the death of more than 1 billion bacterial cells.

4. With the help of neat & labeled diagram describe steps in RDT.

Ans: Steps in r-DNA Technology are:

1. Isolating genomic DNA of a donor.
2. DNA fragmentation with molecular scissor or restriction endonuclease enzyme.
3. Fragments are screened for desired genes.
4. Insertion of fragment into a cloning vector develops r-DNA or chimeric DNA.
5. Introducing r-DNA into competent host cell.
6. Culturing these cells to obtain clones of desired fragment of DNA.
7. Use of these copies to transform host cell so as to express the desired gene.



Ans:

5. What are jumping genes?

Ans: The sequence of DNA that can move or transpose themselves to new position within the genome of a single cell are known as transposons / jumping genes.

6. Explain different biosafety issues which may arise due to genetically modified organisms?(2M)

Ans: Biosafety issues which may arise due to genetically modified organisms are:

1. Cross pollination between GM plants & wild varieties will lead to contamination of gene pools of wild varieties.
2. Consumption of GM food may develop allergies.
3. GM microbes may escape from laboratory & may prove to be hazardous. Therefore manipulation of living organisms needs regulation.
4. Issues related to biopiracy & biopatents may arise.

7. Give the applications of PCR Technique.(3M)(any 6 points)

Ans: Applications of PCR:

1. It is used in DNA cloning.
2. It is used in Gene amplification
3. It is used in DNA based phylogeny or functional analysis of genes
4. It is used in diagnosis of hereditary disorders.
5. It is used in DNA fingerprinting
6. It is used in detection & diagnosis of infectious diseases
7. It is used in cancer diagnosis.

8. Name the enzyme responsible for delay in ripening of tomato fruit.

Ans: The enzyme responsible for delay in ripening of tomato fruit is **“Polygalacturonase”**.

9. Explain types of DNA library.

Ans: DNA library is of 2 types; a) Genomic library b) c-DNA library

a) Genomic library

1. It is collection of all clones of DNA fragments that represent complete genome of an organism.
2. For construction of genomic library of an organism its entire genome is isolated & cut into fragments using suitable restriction endonuclease.
3. These fragments are then inserted into cloning vectors.
4. The recombinant vectors are then transferred into suitable organism such as bacteria/ yeast.
5. One fragment in each host cell & then transformed organisms are cultured to produce their clones & are stored.

b) cDNA library

1. It is complementary DNA.
2. It is produced using mRNA by the process called as reverse transcription.
3. In eukaryotes there is coding & non-coding sequence in the genes.
4. The mRNA carrying transcribed coding sequences is translated into protein.
5. When mRNA of different cells, tissues & organs at different time & phases in the life cycle of an organism are isolated using reverse transcriptase enzyme converted into cDNA.
6. A library having cDNA for each & every type of structure functional protein can be constructed by inserting into suitable vector & then cloning it in a proper host such as *E. coli*.

10. Describe the steps of PCR technique.

Ans: Essential steps are:

- 1) Heat denaturation: Heating of DNA at about 91°C for breaking hydrogen bonds & formation of ssDNA. DNA molecule with more G-C pairs needs higher temperature.
- 2) Annealing: It involves pairing of primers (oligonucleotides) to ssDNA segment at 55°C .
- 3) Polymerisation: Attachment of dNTP (deoxyribose nucleotide tri phosphate) behind the primer on ssDNA, at 72°C in presence of Taq polymerase enzyme.

11. Define RDT, Give the basic steps of RDT & give 3 examples of therapeutic products produced by RDT.

Ans: RDT: Recombinant DNA (rDNA) technology is the technique of manipulating the genome of a cell or organism so as to change the phenotype desirably.

Steps in r-DNA Technology are:

1. **Isolation of genomic DNA of donor:** Isolating genomic DNA of a donor.
2. **Fragmentation:** DNA fragmentation with molecular scissor or restriction endonuclease enzyme.
3. **Screening:** Fragments are screened for desired genes.
4. **Insertion of desired genes:** Insertion of fragment into a cloning vector develops r-DNA or chimeric DNA.
5. **Introduction of r-DNA:** Introducing r-DNA into competent host cell.
6. **Culturing:** Culturing these cells to obtain clones of desired fragment of DNA.
7. **Transformation:** Use of these copies to transform host cell so as to express the desired gene.

Therapeutic products produced by RDT.

1. **Blood Proteins;** erythropoietin factor VII, VIII, IX, tissue plasminogen activator (TPA) urokinase.
2. **Human hormones:** eg: epidermal growth factor, FSH, insulin, nerve growth factor, relaxin, somatotropin.
3. **Immune modulators:** α and β interferon, lysozyme, tumor, necrosis factor, colony stimulation factor.
4. **Vaccines:** eg: Cytomegalovires, hepatitis-B, measles, raies, etc.

(def:1m, Steps: 3M, eg:3m)

1. Give the name of the process involved in DNA fragmentation.

Ans: The process involved in DNA fragmentation is called as **restriction digestion**.

2. Enlist the steps involved in solving disputed parentage by DNA profiling.

Ans: The steps involved in solving disputed parentage by DNA profiling:

- | | |
|----------------------|----------------------|
| a) DNA isolation | e) Southern Blotting |
| b) DNA amplification | f) Hybridisation |
| c) DNA fragmentation | g) Photography |
| d) Eelctrophoresis | |

3. Explain DNA fingerprinting process in detail.

Ans: Def: *The technique developed to identify a person with the help of DNA restriction analysis is known as DNA profiling/ DNA fingerprinting.*

1. In a population every person shows unusual sequence of 20-100 base pairs which are repeated several times. They are termed as **Variable number of Tandem Repeats or VNTR**

2. The length of the region having VNTRs is different in each individual & hence is the key factor in DNA profiling. The steps followed in this technique are as follows:

a) **DNA Isolation:** DNA is obtained from material like blood, semen, hair roots, tissue sample etc.

b) **DNA amplification:** If DNA quantity is small it is subjected to in vitro replication by a technique called PCR (Polymerase Chain reaction) to obtain more copies of DNA

c) **DNA Fragmentation:** The DNA sample is then subjected to restriction end nuclease which cut DNA at specific location making many pieces of DNA having variable length. This phenomenon is called Restriction Fragment Length Polymorphism. This process is called restriction digestion.

D) **Electrophoresis:** DNA samples are loaded on agarose gel electrophoresis. The charge is applied. DNA fragment being negatively charged move towards positive pole. Their movement depends on the length of the fragment. This results in formation of bands.

E) **Southern blotting:** The bands are now blotted on nylon membrane (nitrocellulose paper) The single stranded strand gets embedded in nylon membrane.

F) **Hybridization:** The bands are flooded with radioactive DNA probes. Due to natural affinity sample DNA & probe DNA form double stranded structure. The remaining single stranded DNA probe are washed off.

G) **Photography:** This nylon membrane is kept in contact with X-ray film. The DNA bands due to radioactive probe give photographic image on X-ray film for documentation.

4. **Name the hormones purified from dog's pancreas by Banting & Best.**

Ans: Insulin is the hormones purified from dog's pancreas by Banting & Best.

5. **Name the radioactive probe used in India for DNA fingerprinting.**

Ans: The radioactive probe used in India for DNA fingerprinting is **banded krait minisatellite**.

6. **What is vaccine?**

Ans: Vaccine is an antigenic preparation used to stimulate the production of antibodies & induce immunity against several diseases.

7. **What is restriction digestion.**

Ans: The process by which restriction endonucleases enzyme cuts DNA molecule at specific locations making pieces of DNA of variable length is known as restriction digestion.

8. **Give applications of vaccine.**

Ans: Applications of vaccine are:

1. to stimulate production of antibodies.
2. to induce immunity against several diseases.
3. to protect against diseases.

9. **What is the use of Tissue Plasminogen activator.**

Ans: Tissue Plasminogen activator is used to prevent blood clot or reverse the blood clot.

10. **What is the goal of Human Genome Project.**

Ans: Goal of Human Genome Project :

1. To identify genes associated with genetic characters.
2. To understand the genetic basis of many diseases.
3. To study human genome.
4. Involved in mapping & sequencing of genome
5. To understand gene structure & functions
6. To understand human evolution.

11. **What does HGP stand for?**

Ans: Human Genome Project

12. **Give applications of DNA fingerprinting technique.(2M)**

Ans: DNA fingerprinting technique is used to:

- a) To establish parentage in dispute cases
- b) To settle insurance claims
- c) To study phylogeny
- d) In genomics
- e) In study of agricultural science

f) In forensic sciences.

13. **Define the term genome.**

Ans: The total number of genes present on haploid set of chromosomes is called as genome.

14. **Give the importance of transgenic animals.(2M)(any 2 points with explanation)**

Ans: Importance of transgenic animals

- 1) **Study of diseases:** Many transgenic animals are designed to study how genes contribute to the development of disease. With the help of these animal, investigations of new treatments for human diseases such as cancer, cystic fibrosis, Alzheimer's disease is possible.
- 2) **Vaccine safety:** Transgenic mice are used to test the safety of polio vaccine.
- 3) **Bioreactors:** Transgenic animals are also used as bioreactors to produce pharmacologically important proteins to treat emphysema. A transgenic cow , Rosie produce human protein enriched milk.
- 4) **Clinical safety testing:** Transgenic animals carry genes, which make them more sensitive to toxic substance than other non-transgenic animals. Toxicity testing in such animals allows us to get result in less time.

15. Write down full form of R.F.L.P.

Ans: Restriction Fragment Length Polymorphism is full form of RFLP

16. Illustrate any 3 genes which can be used in gene therapy.

Ans: Genes used in gene therapy are as follows:-

1. Tissue plasminogen activator (TPA) used to prevent or reverse blood clots.
2. Human growth hormone producer gene to treat pituitary dwarfism.
3. Tissue growth Factor beta (TGF- β) promotes new blood vessels & epidermal growth. Useful in wound healing & burns.
4. Human blood clotting factor VIII to treat haemophiliacs
5. Human Insulin (HUMULIN) to treat insulin dependent diabetes.
6. DNAase to treat cystic fibrosis
7. Bovine growth hormone to increase cattle & dairy yield.
8. Recombinant Vaccines for prophylaxis of human & animal viral diseases. (hepatitis B)
9. Genetically engineered bacteria & other micro organisms for improved production of industrial enzymes.

17. What is adjuvant?

Ans: It is the substance which improves the immune response to the toxoids.

18. Who produced the first vaccine and in which year against which disease?

Ans: The first vaccine was produced by Edward Jenner in 1796 against smallpox disease.

19. Describe the process of vaccine production?

Ans: 1. Vaccine is an antigenic preparation used to stimulate the production of antibodies & induce immunity against several diseases.

3. Production of vaccine involves growing a organism in a growth medium to produce large amount of toxins, & then altered to give toxoids which are no longer toxic.
4. The toxoids become antigens which the immune system recognizes & hence produces antibodies to act against genuine toxins.
5. The live organism must be deactivated & the resulting mixture is diluted & mixed with an adjuvant a substance which improves immune response to the toxoids.
6. The first vaccine was produced by Edward Jenner in 1796 in an attempt to provide protection against small pox.

.....